

Answer Sheet - Grade 9 Physical Science - Forces and Motion - Short Test

Q1. An object is moving at a steady speed in a straight line. The net force on it is zero.

What will the object do next?

☐ A ☐ B ☒ C ☐ D

Q2. Which unit is used to measure force?

☐ A ☒ B ☐ C ☐ D

Q3. A 2 kg trolley accelerates at 3 m/s squared. What is the net force on it?

☐ A ☐ B ☒ C ☐ D

Q4. A book rests on a table. Using forces, explain why the book is not accelerating.

The book's weight pulls it down and the table pushes up on it with the normal force. The two forces are equal and opposite so they cancel and the net force is zero, so the book does not accelerate.

Q5. A cyclist rides at a steady speed along a flat, straight road. (a) The forward force from the pedals is ... the total resistive force. Mark one letter.

☐ A ☒ B ☐ C

Q5. (b) Explain your choice in (a).

The cyclist is going at a steady speed so the acceleration is zero. That means the net force is zero, so the pedalling force is the same as the air resistance and friction. This is Newton's first law.